

Report 8 - Grid Model Family for OpenHPL/OpenHPLjl

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Abstract

This report introduces the grid model family. The first implementation is an infinite-grid frequency boundary used to close reduced electromechanical models and prepare the package for SMIB studies.

1 Updated Context

The grid is the downstream electrical boundary:

$$\boxed{\text{Generator} \rightarrow \text{Grid}}.$$

For the first SMIB studies, the grid is treated as infinitely strong.

2 Generalized Concept Model

An infinite grid fixes electrical frequency while accepting whatever active power is injected by the connected generator.

3 Other Models Available in the Literature

A useful hierarchy is

$$\boxed{\text{infinite bus} \rightarrow \text{frequency-dependent grid} \rightarrow \text{SMIB network} \rightarrow \text{multi-machine grid}}.$$

4 Mathematics Involved

For nominal frequency f_0 ,

$$\boxed{\omega_g = 2\pi f_0}.$$

At 50 Hz,

$$\omega_g = 100\pi \text{ rad s}^{-1}.$$

5 OpenHPL-Specific Modeling Interpretation

OpenHPL contains power-system interfaces intended to couple hydropower plant dynamics to electrical-system models. OpenHPLjl first establishes a reduced active-power/frequency port before introducing voltage-angle network equations.

6 Implementation in Julia ModelingToolkit / SciML

```
@connector ElectricalPowerPort begin
    omega(t)
    P(t), [connect = Flow]
end

@component function InfiniteGrid(; name, f0=50.0)
    @named port = ElectricalPowerPort()
    port.omega ~ 2*pi*f0
end
```

7 Model Assembly and Connections

The ideal generator electrical port connects directly to the infinite grid. The connected generator determines active-power exchange while the grid fixes frequency.

8 Numerical Experiment / Validation

For

$$f_0 = 50 \text{ Hz},$$

the expected angular frequency is

$$\omega_g = 100\pi \approx 314.159 \text{ rad s}^{-1}.$$

9 Expected Outputs

Useful outputs are

$$f(t), \quad \omega(t), \quad P(t).$$

For the infinite grid, frequency remains constant.

10 Engineering Interpretation

The infinite-grid model is a boundary condition, not a dynamic grid. It is appropriate for early SMIB validation before finite grid inertia, load damping, transmission, and inter-area dynamics are introduced.

Conclusion

The first electrical boundary is now available. The next milestone is the complete reduced chain

$$\text{Turbine} \rightarrow \text{Shaft} \rightarrow \text{Generator} \rightarrow \text{InfiniteGrid}$$

followed by a full hydropower SMIB model.